

Speech and language technology (DA 3452)

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Lecture 1 – July 28
Introduction to the course

Course Details

Title : Speech and Language Technology

Course No : DA3452

Slot : E Slot

Timings : Tue 11-12, Wed 10-11, Thu 8-9, (Fri 5-6)

Instructor : Balaji Srinivasan, Room 654, NAC-2, Email : sbalaji@dsai.iitm.ac.in

TAs : Sridhar Ramachandran, Rishi Mishra, Others to be decided

Course Objective:

Understand how human language (written or spoken) can be **modeled**.

That is, converted into computational objects, recognized, learned from, generated, retrieved, and evaluated.

Course Organization (6 Modules)

Understand how computers can represent, understand, learn from, generate, and interact using human language.

1. Representing Language Computationally
2. Predicting Language and Linguistic Labels
3. Learning and Generating Sequences
4. Foundation Models and Adaptation
5. Knowledge-Grounding, Evaluation and Robustness
6. Speech Representation, Recognition and Synthesis

What this course is and is not

This course is not primarily about

1. Deep Learning (You have a parallel course)
2. Software Tools
3. AI Products
4. Prompt Engineering!

This is an academic course on

1. Representing language computationally
2. Building models that learn and generate language
3. Designing, evaluating and grounding language technologies
4. Understanding modern speech and language systems

Reference Material

Books —

1. **Speech and Language Processing – Jurafsky and Martin (available draft online)**
2. **Build a Large Language Model (from Scratch) – Sebastian Raschka**

Supplementary Resources

1. **Stanford CS224N (NLP with Deep Learning)**
2. **Stanford CS224S (Spoken Language Processing)**
3. **Hugging Face LLM Course**
4. **Research Papers**

Evaluation Method

Warning: This is tentative

This will depend on the number of students in the class as well as the number of TAs assigned

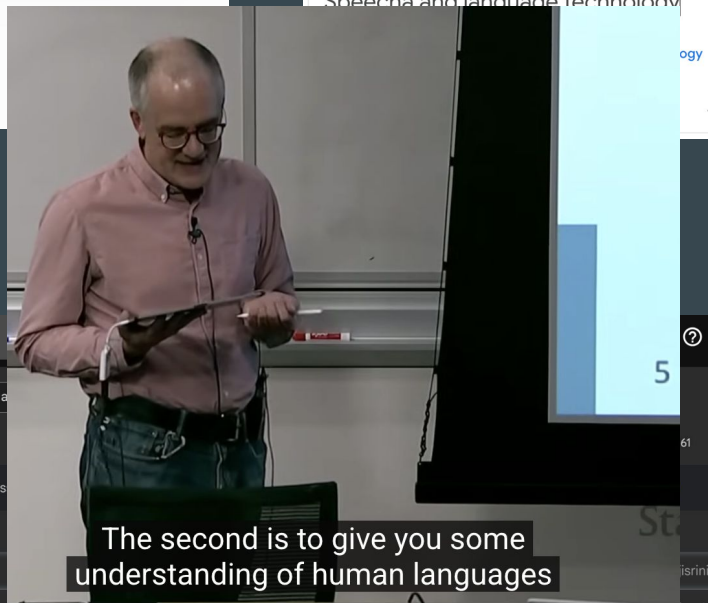
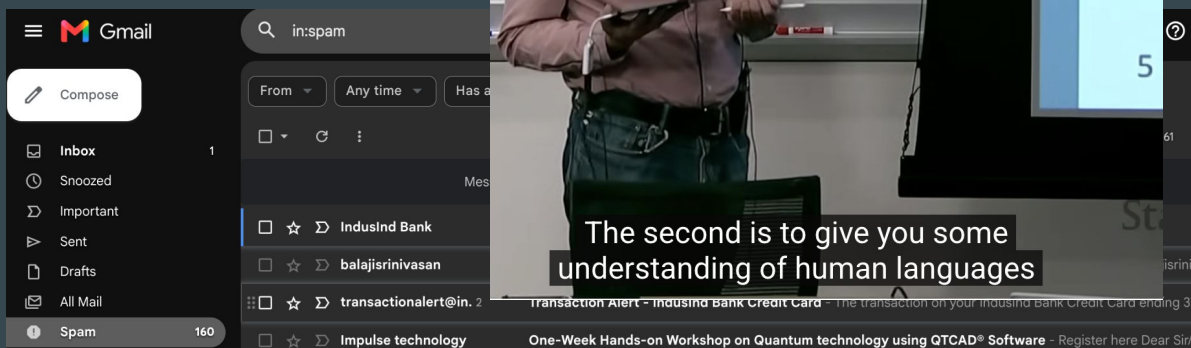
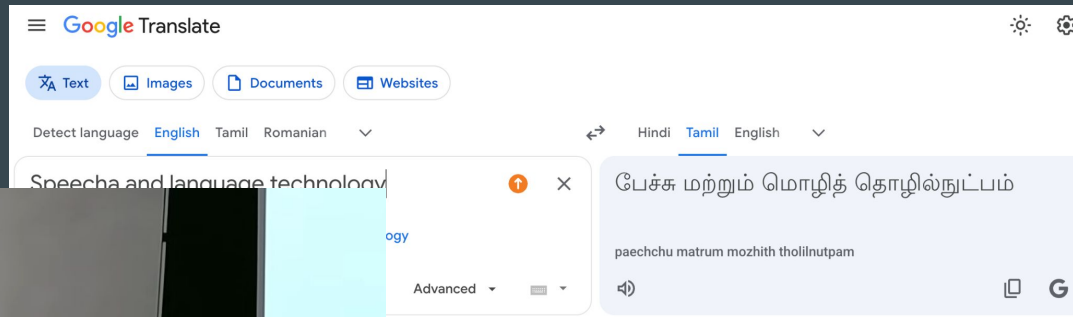
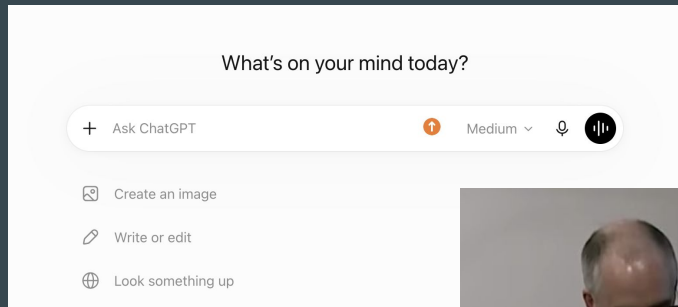
Approximately

- 6 Module Tests – 50%
- End Semester – 50%

Module 1

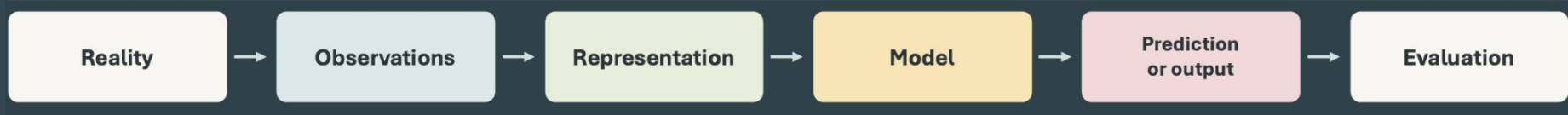
Representing Language Computationally

Speech and Language Models are everywhere



What does it mean to model?

A model is a deliberately simplified representation of some aspect of reality, built for a purpose.



**A model is not a complete representation of reality.
It keeps what matters for a task and discards what does not.**

Modeling – Physics vs Language

Modelling in physics

- Choose state variables
- Make simplifying assumptions
- Specify or discover relationships
- Estimate parameters
- Predict observations
- Compare with experiments
- State the domain of validity

Modelling language

- Choose linguistic or computational units
- Choose context and representation
- Learn statistical relationships
- Train parameters
- Predict words, labels, or sequences
- Evaluate on held-out data
- Study domain shift and failure modes

Side quest – Can LLMs model physics?

Google DeepMind

Jan 27th, 2026

LLMs can't jump

Tom Zahavy, Google DeepMind

How do we fundamentally discover new things? In a letter to Maurice Solovine, Albert Einstein conceptualized discovery as a cyclical process involving an intuitive 'jump' from sensory experience to axioms, followed by logical deduction. While Generative AI has mastered Induction (statistical pattern matching) and is rapidly conquering Deduction (formal proof), we argue it lacks the mechanism for Abduction—the generation of novel explanatory hypotheses. Using Einstein's formulation of General Relativity as a computational case study, we demonstrate that the prevailing theory of "creativity as data compression" (induction) fails to account for discoveries where observational data is scarce. This position paper argues that while a modern Large Language Model could plausibly execute the deductive phase of proving theorems from established premises, it is structurally incapable of the abductive 'Jump' required to formulate those premises. We identify the translation of simulation into formal axioms as the critical bottleneck in artificial scientific invention, and propose that physically consistent, multimodal world models offer the necessary sensory grounding to bridge this divide.

1. Introduction

What characterizes the cognitive leap required for scientific invention? A prevailing view in the AI community, notably championed by Schmidhuber (2008), suggests that scientific discovery is fundamentally a problem of compression—the search for a simple program that concisely explains observations. This view implicitly frames discovery as *Induction*: inferring general rules from observations based on statistical frequency. Concurrently, the success of systems like AlphaProof (Hubert et al., 2025) in Olympiad-level mathematics suggests that AI is mastering *Deduction*: the formal derivation of theorems from established premises.

If scientific discovery were merely the sum of these two parts, modern Large Language Models (LLMs) should theoretically be capable of inventing theories like General Relativity given sufficient compute.

In this paper, we challenge this reductionist premise by treating Albert Einstein's formulation of General Relativity as a computational case study. Adopting Einstein's cyclical model of invention—illustrated in Fig. 1—we map the process from Sense Experience (*E*) to a System of Axioms (*A*) via a conceptual Jump (*J*), followed by the deduction and verification of theorems. While we concede that a modern LLM could plausibly perform the deductive work if initialized with Einstein's assumptions, the formulation of the axioms remains the bottleneck.

By reconstructing the historical context in Section 2, we show that the scarcity of experimental data precludes induction as a viable explanation. General Relativity also

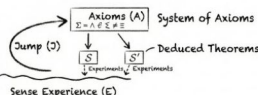


Figure 1 | A generative AI reconstruction of Einstein's E-J-A diagram. Einstein drew this diagram in a letter to Maurice Solovine, showing a cyclical line jumping from Sense Experience (*E*) to Axioms (*A*) via a Jump (*J*), and then deducing logical consequences. Ironically, the hallucination of the axiomatic symbols highlights the very difficulty of automating the jump.

beyond induction and deduction: Abduction.

To formalize this, we adopt the framework of Peirce (1934), which categorizes inference based on the structural permutation of a *Rule* (function definition), a *Case* (input), and a *Result* (return value):

- **Deduction** ($\text{Rule} + \text{Case} \rightarrow \text{Result}$) is the analytic application of a Rule to a Case to predict a Result. It is the only mode that guarantees truth (e.g., executing code to verify output).
- **Induction** ($\text{Case} + \text{Result} \rightarrow \text{Rule}$) is the synthetic derivation of a Rule from the accumulation of Cases and Results. It validates hypotheses through statistical frequency (e.g., generating a function to satisfy unit tests).

Why language is difficult to model

Language is harder to model than physics

1. Its units are neither natural nor universal
2. Ambiguity is very common
3. Language varies across speakers, genres and over time
4. Context can change meaning
5. Several outputs may be acceptable for the same input
6. The underlying objects and process to be modeled are themselves unclear!

How many words in this sentence?

“They picnicked by the pool, then lay back on the grass and looked at the stars.”

16

if punctuation is excluded

18

if punctuation is counted

The answer depends on the representation and the task.